Jay A. Panchal

Software Engineer

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Summary

Software Engineer with 1.5+ years of experience building backend systems and REST APIs with Java, Python, and Node.js, now completing a Master of Science in Computer Science at Loyola Marymount University.

Education

Loyola Marymount University — Master of Science in Computer Science (MS) *Los Angeles, CA | Expected Dec 2026*
GPA: 3.88 | *Coursework*: Algorithms, Database Systems, Machine Learning, Deep Learning, Generative AI

Gujarat Technological University — Bachelor of Engineering in Computer Engineering (BE) *Ahmedabad, India | May 2023*

Professional Experience

Teaching Assistant, Machine Learning — Loyola Marymount University *Los Angeles, CA | Aug 2025 - Present*

- Sole Teaching Assistant for the graduate Machine Learning course; grade 5 written and coding assignments with clear, consistent feedback; previously supported Deep Learning

Software Engineer — Aequitas Infotech Pvt. Ltd. *Ahmedabad, India | Jan 2024 - Aug 2024*

- Maintained and improved backend features in Java and MySQL for a practice management platform used by tens of thousands of Chartered Accountants across India
- Built custom features for individual clients and fixed client-reported bugs, often during live support calls
- Wrote and updated SQL queries to support new features and fix data issues
- Gathered requirements in client meetings and trained users on the platform in sessions of up to 300 people

Backend Developer — CodeHabit *Gandhinagar, India | Sep 2023 - Dec 2023*

- Built REST APIs with Node.js, Express, and MongoDB for a hospital management system used by 25+ staff, covering patient records and staff scheduling
- Implemented JSON Web Token (JWT) authentication and role-based access control for staff, admin, and patient users
- Tested all API endpoints in Postman before each release

Software Engineer Intern — Praxware Technologies *Ahmedabad, India | Jan 2023 - Aug 2023*

- Built a sales and inventory web application with Python, Django, and MySQL, including REST APIs, an HTML and Bootstrap interface, and create, read, update, and delete (CRUD) features

Software Engineer Intern — HN Techno *Ahmedabad, India | Jun 2022 - Jul 2022*

- Built an inventory management web application with Angular, TypeScript, and JavaScript, using reusable components and CRUD features for stock tracking

Projects

GoTinerary — AI Trip Planner | React, FastAPI, Python, Gemini API, Google Places API GitHub

- Built solo in one weekend a full-stack web app with a React front end and a FastAPI back end that plans trips from a user's request
- Used Gemini tool calling so the model queries the Google Places API on its own to find real places, and connected the back end to the front end through REST API endpoints

atlas.db — Database Benchmarking System | Python, PostgreSQL, MongoDB, AWS RDS

- Benchmarked PostgreSQL against MongoDB on 622K Los Angeles traffic collision records, loading raw CSV data into 4 linked tables on AWS RDS
- PostgreSQL won 4 of 5 analytical queries, running a JOIN-heavy query in 1.72s versus 92.64s; won Most Cohesive System Architecture and Most Effective Presentation awards

Technical Skills

- **Languages**: Java, Python, JavaScript, TypeScript, SQL, C, C++
- **Back-End**: Node.js, Express, Django, FastAPI, REST APIs
- **Front-End**: Angular, React, HTML, CSS, Bootstrap
- **Databases and Cloud**: MySQL, MongoDB, PostgreSQL, AWS (RDS)
- **Tools**: Git, GitHub, Postman, Eclipse

Awards

- **LA Hacks 2026 — Best Use of Gemini API (MLH)** | Oct 2026: Won for GoTinerary, built solo in one weekend
- **LMU Datathon — 3rd Place ($1,000)** | Mar 2026: Analyzed the NLSY97 dataset with Python, pandas, and regression
- **LMU BEST Entrepreneurship Bootcamp — 3rd Place** | Jan 2026: Pitched an AI-powered travel app in a 4-person team